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Current Perspective

Unlocking clinical quantum oncology through quantum control

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ABSTRACT

Quantum technologies present a transformative frontier for oncology, promising significant advancements in diagnostics, treatment precision, and drug discovery. The clinical realization of this potential is fundamentally reliant on mastering quantum control—the precise manipulation of quantum systems. This perspective details how quantum control is integral across emerging quantum oncology applications, potentially enhancing quantum imaging and quantum sensing by boosting signal sensitivity, improving clarity (e.g., fivefold Signal to Noise Ratio increase), and enabling subcellular visualization. In quantum omics, it could facilitate precise single-cell analysis and drive quantum machine learning for advanced biomarker discovery and validation. For treatment, such precise manipulation could optimise radiation targeting in quantum radiotherapy, could enable real-time tracking for precise quantum surgery, guide targeted delivery in quantum chemotherapy via controlled nanoparticle release, and could enhance the design and precision of quantum immunotherapy and personalised therapies. Overcoming challenges such as coherence and scalability is necessary, but advancements in quantum control are paramount for translating these capabilities into clinical benefits, ultimately improving cancer diagnosis and patient outcomes.

1. Introduction

Oncology could be transformed by quantum technologies, offering significant improvements in diagnostics, treatment precision, and drug discovery [1]. Building on our previous broader review of quantum technologies in oncology, this current perspective emphasizes the critical need for quantum control based on the current state of the art. Quantum-enhanced technologies for cancer care and research will require a multidisciplinary approach to integrate into clinical practice, combining the latest scientific discoveries with practical, real-world applications. A growing momentum is evident, with 32 active clinical trials exploring 25 distinct quantum healthcare innovations (sourced from clinicaltrials.gov as of June 1, 2025), underscoring the growing momentum in this field. These trials are predominantly in early phases (Phase I/II) and largely focus on quantum-enhanced imaging and diagnostic algorithms, with a growing number of studies exploring novel therapeutic approaches, including those in quantum radiotherapy and targeted drug delivery.

In classical physics, a system's properties are always definite and observable; for example, a ball has a specific position and momentum at all times. In contrast, quantum mechanics governs matter and energy at atomic and subatomic levels. Here, particles exist in unique "quantum states" where, before measurement, they can be a combination of all possible states simultaneously – a phenomenon known as superposition [1]. This inherent multi-state and probabilistic nature, which then "collapses" into a definite state only upon measurement, underpins the distinct power of quantum computing and sensing [1]. This enables processing information fundamentally differently, allowing systems to handle multiple possibilities in parallel, detect subtle changes with extreme precision, and establish complex correlations within data. These unique capabilities provide computational and sensing advantages beyond traditional methods, forming the foundation for quantum technologies.

Central to harnessing these phenomena and realizing the full potential of these early-phase clinical advancements is mastering quantum states control [1]. Quantum control —the precise manipulation of particle states at the smallest scale – is not merely an optimization technique; it forms a new foundation for all quantum-based oncology technologies. Indeed, this precise control is crucial for improving the precision of robotic surgical procedures, guiding drug delivery directly to cancer cells, and providing more accurate diagnostics by manipulating biological molecules at a very small scale. As this technology evolves, advances in quantum control could play a key role in personalised medicine, allowing for therapies customised to the unique

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characteristics of individual patients. Additionally, integrating quantum control with artificial intelligence (AI) could also help predict disease progression and treatment response in real time, ultimately improving patient outcomes. This merger of quantum control and AI, specifically leveraging the potential of quantum machine learning (QML), promises dramatic impacts. QML aims to speed up and improve classical machine-learning models and could learn to recognize patterns in data that its classical counterparts would miss. Its ability to process data, spot patterns, and make inferences by learning rules from unfamiliar situations will be crucial for downstream clinical applications such as auto-contouring for radiotherapy (RXT), advanced image analysis and interpretation, data integration, and algorithmically driven multidisciplinary team (MDT) decisions. However, more research and development in this domain are needed to integrate these techniques into this wide array of settings and clinical challenges.

2. Quantum imaging and diagnostics

Quantum technologies, driven by precise quantum manipulation, are already being integrated into cancer imaging and diagnostics, to make them more precise and sensitive. Quantum-enhanced mammography, using techniques such as single-photon detection to capture fainter signals, has shown a 20 % increase in sensitivity, improving detection rates from 75 % to 95 % in preclinical studies [1,2]. This means that tumours as small as 1–2 mm, compared to 4–5 mm in standard mammography, can now be detected, potentially leading to earlier diagnosis and better survival rates. This improvement comes from the ability to detect fainter signals, providing a more precise measurement of breast tissue density. This enhanced sensitivity, enabled by quantum control, allows for earlier detection of breast cancer, crucial in improving patient outcomes. However, the clinical significance of detecting increasingly smaller lesions, particularly regarding the potential for overdiagnosis of indolent cancers, necessitates rigorous prospective validation. Similarly, quantum-enhanced MRI has dramatically improved scan clarity by amplifying signal strength. A technique known as quantum hyperpolarisation aligns atomic particles to boost the signal-to-noise ratio (SNR) fivefold [3,4], allowing for earlier detection of tumours and more accurate cancer staging. While promising for early detection, the clinical implications of identifying increasingly subtle molecular changes require thorough investigation to ensure benefits outweigh risks, including overdiagnosis. This advancement enables clinicians to observe subtle molecular changes in tumours that would otherwise go undetected in standard MRI scans, providing valuable information for planning treatment. Moreover, for CT scans, photon-counting detectors enhanced with quantum technology have improved contrast-to-noise ratios by 33.3 %, from 15 to 20, providing sharper images while reducing radiation exposure [1]. This is particularly beneficial for paediatric oncology patients, who are more vulnerable to radiation, minimising their long-term risk from imaging procedures. Finally, Quantum-enhanced fluorescence microscopy and ghost imaging allow visualization at the subcellular level [1], potentially detecting malignancies at their earliest stages by visualising specific cancer-related biomarkers (e.g., HER2/neu for breast cancer, PSA for prostate cancer, and CA-125 for ovarian cancer) [5,6], enabling earlier intervention and potentially improving patient outcomes. This precise manipulation over quantum light enables visualization of cancer cells with unprecedented resolution, opening new frontiers in early intervention and treatment planning.

3. Molecular-level cancer detection

At the molecular level, quantum control is enabling unprecedented insights into cancer biology. Quantum-controlled microfluidics, combined with advanced tools such as mass spectrometry and NMR spectroscopy, are enabling more detailed analysis of single cancer cells and the variations within tumours. Precise manipulation of fluids at the nanoscale, enabled by quantum control, allows for the isolation of individual cancer cells, revealing subtle variations in tumour properties that would otherwise be masked in bulk tissue analysis [1]. This quantum-enhanced ability to analyse single cells has the potential to lead to more targeted treatments and a better understanding of how cancers grow and spread through a new era in (sub) molecular pathology. For the clinical setting, especially pathology, this means being able to analyse the unique characteristics of individual cancer cells within a tumour, which can help predict how the tumour will respond to treatment and identify potential mechanisms of drug resistance. Furthermore, quantum machine learning (QML) algorithms, leveraging quantum control for optimised data processing, are assisting in the wider biomarker discovery and validation. QML takes advantage of quantum properties (like superposition) to process vast amounts of genomic and imaging data quickly, unveiling insights critical for personalised cancer treatments that might not be detected by traditional methods [1]. The precise control of quantum technology plays a crucial role in optimising the performance of QML algorithms, enabling them to identify complex patterns and correlations that are essential for developing targeted therapies. However, substantial effort is still required to develop robust models that can meaningfully integrate vast amounts of highly granular data and translate them into clinically relevant and actionable outputs. This enhanced data analysis capability could help identify new clinically relevant biomarkers for diagnosis, prognosis and prediction.

4. Advancing cancer treatment precision

Quantum technologies, leveraging the power of quantum control, are showing great promise in improving cancer across a range of treatment modalities. In robotic surgery, quantum-controlled sensors, for example, can help track tumours in real time, enabling dynamic adjustments to the surgical instrument manipulation during procedures. This allows for more precise tumour resection, minimising damage to surrounding healthy tissue [1]. For patients, this could translate to less invasive surgery, faster recovery times, and reduced risk of complications. In radiotherapy, quantum control algorithms are being used to optimise radiation beam placement. These algorithms, enabled by precise manipulation of beam parameters [1], explore numerous possible beam configurations simultaneously, ensuring that radiation is precisely targeted at tumours while sparing healthy tissues. This quantum-assisted approach, leveraging quantum control for enhanced precision, can reduce side effects by 30 % and enhance radiation therapy's effectiveness [1,7]. This improved precision in radiotherapy means that patients can receive higher doses of radiation to the tumour while minimising damage to healthy tissues, leading to better tumour control and fewer side-effects. Quantum-controlled nanoparticles, such as quantum dots and NV diamond nanoparticles, are being used for targeted drug delivery. Quantum control over the release mechanisms of these nanoparticles allows for chemotherapy to be delivered more directly to cancer cells, significantly reducing systemic toxicity [1,8]. These controlled nanoparticles, through advanced quantum methods [1], can also improve the effectiveness of radiation therapy, further boosting treatment outcomes. Targeted drug delivery, enabled by quantum control, can significantly reduce the side effects of chemotherapy and radiotherapy by delivering the drugs directly to the tumour cells and minimising exposure to healthy tissues.

5. Personalised medicine and quantum control

Quantum technologies are poised to integrate into new advances with cancer immunotherapy, with quantum control playing a crucial role in each stage of development. Quantum machine learning (QML), enhanced by quantum control techniques for optimised data analysis, is enabling personalised treatment selection by analysing patient data [1]. The principles of quantum control are also being applied in quantum chemistry techniques used to design more effective and targeted immunotherapies, allowing for precise manipulation of molecular interactions and improved drug design [1]. Furthermore, this advanced control is directly enhancing the precision and safety of these therapies, including gene-editing tools, by enabling fine-tuned manipulation of molecular processes [1]. These combined, quantum-control-driven approaches have the potential to significantly improve outcomes for patients undergoing immunotherapy [1]. By tailoring immunotherapies to the specific characteristics of each patient's tumour and immune system, quantum control can help to improve the effectiveness of these treatments and reduce the risk of adverse events.

6. Challenges and barriers

Despite these promising advances, challenges remain in integrating quantum technologies into clinical oncology. The coherence times (the duration a quantum system can remain stable) of quantum sensors must be extended [9,10]. Currently, these systems last only about 50–60 microseconds, but the goal is to extend them to more than 100 ms for clinical use [1]. Longer coherence times would allow real-time monitoring of tumour progression and drug delivery. Scalability also presents a challenge. For clinical applications, quantum computers need to simulate 1000–10,000 qubits (quantum bits) to accurately model the tumour microenvironment and predict treatment outcomes [11]. Current quantum computers can simulate only about 50–100 qubits, far short of the target required for real-world clinical applications [12]. Researchers are actively working to overcome these challenges by developing new materials and quantum control techniques to extend coherence times and increase the number of qubits [13–15]. As quantum technologies progress, ethical considerations must be addressed. Data privacy and algorithmic fairness are critical issues, requiring robust anonymisation techniques, secure data storage, and transparent AI algorithms to ensure equity in healthcare access. In addition, regulatory frameworks need to be developed to ensure these technologies are rigorously tested and validated, demonstrating real clinical improvements, such as increased diagnostic accuracy and treatment efficacy. Beyond technical and ethical hurdles, the significant cost implications of developing and deploying quantum technologies represent a substantial barrier. Ensuring these innovations are not only effective but also economically feasible and equitably accessible will be crucial for widespread clinical adoption. A significant remaining concern is about how highly focused, analytical and reductive explorations (often enabled by AI and quantum methods) translate to the holistic, complex adaptive system of cancer, both biologically and socio-ecologically.

7. Conclusion

The potential of quantum technologies in oncology is vast, but to achieve clinical impact, quantum control, the ability to manipulate quantum systems with high precision, is essential. As quantum control improves, it will enable accurate diagnostics, personalised therapies and targeted treatments for patients with cancer. Indeed, the applications discussed in this perspective implicitly illustrate a progression, from current advancements in quantum imaging and sensing that are closer to clinical adoption, to future potentials in areas like highly personalised treatments, outlining a timeline from proximate to distant impact. This will require collaboration among clinicians, physicists and engineers, ultimately offering better survival outcomes, improved quality of life, and more effective cancer treatments for patients (Fig. 1).

CRediT authorship contribution statement

Arnie Purushotham: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Matarèse Bruno F. E.:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization.

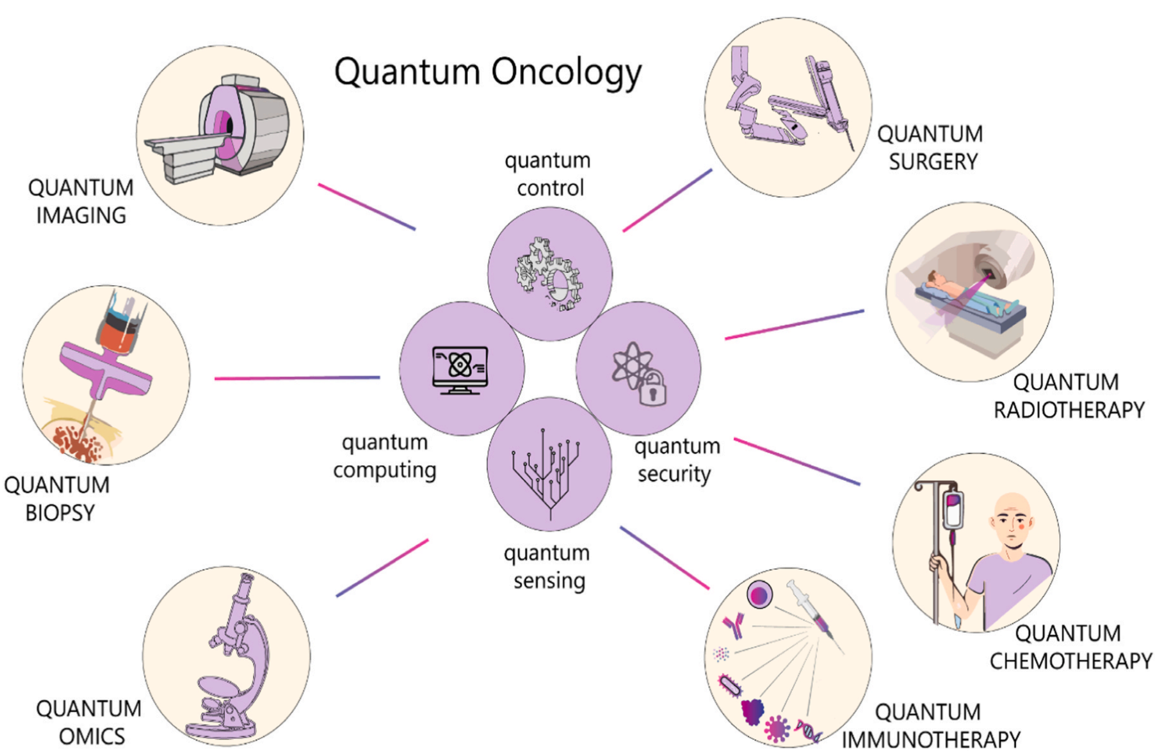

Fig. 1. Overview of quantum oncology applications in diagnostics and therapy. This figure provides an overview of quantum oncology applications in diagnostics (quantum imaging, quantum biopsy, and quantum omics) and therapy (quantum surgery, quantum radiotherapy, quantum chemotherapy, and quantum immunotherapy).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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